

EX-1: Load a real-world dataset (CSV/JSON) into a Pandas DataFrame and inspect its structure using `.head()`, `.info()`, `.shape`, and `.describe()`.

E-COMMERCE Website DATA Set.

Aim

To load the May 2022 E-Commerce Product Pricing dataset (CSV) into a Pandas DataFrame and examine its structure using the `.head()`, `.info()`, `.shape`, and `.describe()` methods.

Algorithm

Step 1: Start the program.

Step 2: Import the **Pandas** library, which is a Python library used for data manipulation and analysis.

Step 3: Read the **E-commerce CSV dataset** using the `pd.read_csv()` function, which loads the CSV file into a **DataFrame** (a two-dimensional table with rows and columns).

Step 4: Display the first five records of the dataset using the `.head()` method to get a quick overview of the data.

Step 5: Use the `.info()` method to view information about the dataset, such as column names, data types, and the number of non-null values.

Step 6: Use the `.shape` attribute to display the dimensions of the dataset (number of rows and columns).

Step 7: Use the `.describe()` method to generate summary statistics such as count, mean, standard deviation, minimum, maximum, and quartile values for numerical columns.

Step 8: Analyze the displayed information to understand the structure and characteristics of the E-commerce dataset.

Step 9: Stop the program.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
153	151 Os230_14	Os230_14	Rozana	Kurta Set	0.4	590	2378	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495
154	152 Os230_14	Os230_14	Rozana	Kurta Set	0.4	590	2378	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495
155	153 Os230_14	Os230_14	Rozana	Kurta Set	0.4	590	2378	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495
156	154 Os230_14	Os230_14	Rozana	Kurta Set	0.4	590	2378	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495
157	155 Os230_14	Os230_14	Rozana	Kurta Set	0.4	590	2378	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495
158	156 Os225_14	Os225_14	Rozana	Kurta Set	0.4	590	2378	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495
159	157 Os225_14	Os225_14	Rozana	Kurta Set	0.4	590	2378	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495
160	158 Os225_14	Os225_14	Rozana	Kurta Set	0.4	590	2378	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495
161	159 Os225_14	Os225_14	Rozana	Kurta Set	0.4	590	2378	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495
162	160 Os225_14	Os225_14	Rozana	Kurta Set	0.4	590	2378	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495
163	161 Os225_14	Os225_14	Rozana	Kurta Set	0.4	590	2378	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495
164	162 Os229_14	Os229_14	Rozana	Kurta Set	0.4	590	2378	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495
165	163 Os229_14	Os229_14	Rozana	Kurta Set	0.4	590	2378	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495
166	164 Os229_14	Os229_14	Rozana	Kurta Set	0.4	590	2378	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495
167	165 Os229_14	Os229_14	Rozana	Kurta Set	0.4	590	2378	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495
168	166 Os229_14	Os229_14	Rozana	Kurta Set	0.4	590	2378	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495
169	167 Os229_14	Os229_14	Rozana	Kurta Set	0.4	590	2378	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495	2495
170	168 Os200_50	Os200_50	Colors-7	Kurta	0.3	381	1538	1595	1595	1595	1595	1595	1595	1595	1595	1595	1595	1595
171	169 Os200_50	Os200_50	Colors-7	Kurta	0.3	381	1538	1595	1595	1595	1595	1595	1595	1595	1595	1595	1595	1595

EDA-Ex-1.csv

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
172	170 Os200_50	Os200_50	Colors-7	Kurta	0.3	381	1538	1595	1595	1595	1595	1595	1595	1595	1595	1595	1595	1595
173	171 Os200_50	Os200_50	Colors-7	Kurta	0.3	381	1538	1595	1595	1595	1595	1595	1595	1595	1595	1595	1595	1595
174	172 Os200_50	Os200_50	Colors-7	Kurta	0.3	381	1538	1595										

Procedure

1. Open the **Kaggle** website and search for a suitable **E-Commerce Product Pricing Dataset**.
2. Download the dataset in **CSV (Comma-Separated Values)** format.
3. Open the downloaded dataset and verify the records and column names.
4. Select the required number of records (198 rows) from the original dataset for easy analysis and save it as a new CSV file.
5. Open **Jupyter Notebook**, **Google Colab**, or any Python IDE.
6. Import the **Pandas** library using the `import pandas as pd` statement.
7. Load the CSV dataset into a Pandas DataFrame using the `pd.read_csv()` function.
8. Verify that the dataset has been loaded successfully.
9. Perform Exploratory Data Analysis (EDA) operations such as `.head()`, `.info()`, `.shape`, and `.describe()`.
10. Observe and analyze the output obtained from each operation.

Python Program :

Program 1: Load the Dataset

Program

```
# Import Pandas library
import pandas as pd

# Load the dataset
df = pd.read_csv("EDA-Ex-1.csv")

# Display the dataset
print(df)
```

Output:

```
IDLE Shell 3.13.1
File Edit Shell Debug Options Window Help
Python 3.13.1 (tags/v3.13.1:0671451, Dec 3 2024, 19:06:28) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: D:/EDA/EX-1-LOAD.PY =====
   index  Sku      Style Id  ... Myntra MRP  Paytm MRP  Snapdeal MRP
0      0  Os206_3141_S  Os206_3141  ...      2295      2295      2295
1      1  Os206_3141_M  Os206_3141  ...      2295      2295      2295
2      2  Os206_3141_L  Os206_3141  ...      2295      2295      2295
3      3  Os206_3141_XL  Os206_3141  ...      2295      2295      2295
4      4  Os206_3141_2XL  Os206_3141  ...      2295      2295      2295
...     ...      ...      ...      ...      ...      ...
194    194  Os198_5045_L  Os198_5045  ...      1595      1595      1595
195    195  Os198_5045_XL  Os198_5045  ...      1595      1595      1595
196    196  Os198_5045_2XL  Os198_5045  ...      1595      1595      1595
197    197  Os198_5045_3XL  Os198_5045  ...      1595      1595      1595
198    198  Os199_5046_S  Os199_5046  ...      1595      1595      1595

[199 rows x 17 columns]
>>>|
```

Program 2: Display the First Five Records (.head())

Program

```
# Import Pandas library
import pandas as pd

# Load the dataset
df = pd.read_csv("EDA-Ex-1.csv")

# Display the first five records
print(df.head())
```

Output:

```
>>> .
===== RESTART: D:/EDA/EX-1-DISPLAY-FIRST5.PY =====
   index      Sku      Style Id  ... Myntra MRP Paytm MRP  Snapdeal MRP
0      0  Os206_3141_S  Os206_3141  ...      2295      2295      2295
1      1  Os206_3141_M  Os206_3141  ...      2295      2295      2295
2      2  Os206_3141_L  Os206_3141  ...      2295      2295      2295
3      3  Os206_3141_XL  Os206_3141  ...      2295      2295      2295
4      4  Os206_3141_2XL  Os206_3141  ...      2295      2295      2295

[5 rows x 17 columns]
>>>
```

Program 3: Display the Last Five Records (.tail())

Program

```
# Import Pandas library
import pandas as pd

# Load the dataset
df = pd.read_csv("EDA-Ex-1.csv")

# Display the last five records
print(df.tail())
```

Output:

```
>>> .
===== RESTART: D:/EDA/EX-1-DISPLAY-LAST5.PY =====
   index      Sku      Style Id  ... Myntra MRP Paytm MRP  Snapdeal MRP
194    194  Os198_5045_L  Os198_5045  ...      1595      1595      1595
195    195  Os198_5045_XL  Os198_5045  ...      1595      1595      1595
196    196  Os198_5045_2XL  Os198_5045  ...      1595      1595      1595
197    197  Os198_5045_3XL  Os198_5045  ...      1595      1595      1595
198    198  Os199_5046_S  Os199_5046  ...      1595      1595      1595

[5 rows x 17 columns]
>>>
```

Program 4: Display Dataset Information (.info())

Program

```
# Import Pandas library
import pandas as pd

# Load the dataset
df = pd.read_csv("EDA-Ex-1.csv")

# Display dataset information
df.info()
```

Output:

```
>>> ===== RESTART: D:/EDA/EX-1-DATASET-INFO.py =====
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 199 entries, 0 to 198
Data columns (total 17 columns):
 #   Column                Non-Null Count  Dtype  
---  --
 0   index                 199 non-null   int64  
 1   Sku                   199 non-null   object  
 2   Style Id              199 non-null   object  
 3   Catalog               199 non-null   object  
 4   Category              199 non-null   object  
 5   Weight                199 non-null   float64 
 6   TP                    199 non-null   int64  
 7   MRP Old               199 non-null   int64  
 8   Final MRP Old         199 non-null   int64  
 9   Ajio MRP              199 non-null   int64  
10   Amazon MRP            199 non-null   int64  
11   Amazon FBA MRP        199 non-null   int64  
12   Flipkart MRP          199 non-null   int64  
13   Limeroad MRP          199 non-null   int64  
14   Myntra MRP            199 non-null   int64  
15   Paytm MRP             199 non-null   int64  
16   Snapdeal MRP          199 non-null   int64  
dtypes: float64(1), int64(12), object(4)
memory usage: 26.6+ KB
>>>
```

Program 5: Display Dataset Dimensions (.shape)

Program

```
# Import Pandas library
import pandas as pd

# Load the dataset
df = pd.read_csv("EDA-Ex-1.csv")

# Display the number of rows and columns
print(df.shape)
```

Output:

```
>>> (199, 17)
```

Program 6: Display Statistical Summary (.describe())

Program

```
# Import Pandas library
import pandas as pd

# Load the dataset
df = pd.read_csv("EDA-Ex-1.csv")

# Display statistical summary
print(df.describe())
```

Output:

```
===== RESTART: D:/EDA/EX-1-SATISTICAL.PY =====
count      index      Weight      ...      Paytm MRP      Snapdeal MRP
mean      99.000000      0.324121      ...      2119.623116      2119.623116
std        57.590508      0.042889      ...      310.012599      310.012599
min         0.000000      0.300000      ...      1595.000000      1595.000000
25%        49.500000      0.300000      ...      1895.000000      1895.000000
50%        99.000000      0.300000      ...      2195.000000      2195.000000
75%       148.500000      0.300000      ...      2295.000000      2295.000000
max       198.000000      0.400000      ...      2495.000000      2495.000000

[8 rows x 13 columns]
>>>
```

Result

Thus, the May 2022 E-Commerce Product Pricing dataset was successfully loaded into a Pandas DataFrame, and its structure, dimensions, and statistical summary were analyzed using Pandas functions.

Exploratory Data Analysis (EDA)